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DATA DESCRIPTOR

New Release of User-Captured Images from the Oregon Health & Science University Melanoma MoleMapper Project

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We announce the release of the OHSU MoleMapper Smartphone Skin Images dataset which contains over six years of new data acquired from the Oregon Health & Science University's (OHSU) MoleMapper study. This released dataset includes 27,499 mole images curated to exclude images with protected health identifiers, 7,305 images of skin patches near the mole images, 1,000 contextual images, and basic metadata from the participants. This data is available to qualified researchers on Sage Bionetwork's Synapse platform under Synapse ID syn51520810 and represents the largest publicly available dataset of consumer-collected smartphone images of pigmented skin lesions. We release these data to the biomedical research community to enable quantitative analysis of images of non-clinician smartphone photography of skin lesions as well as to better understand what lesions appear concerning to the public. These data are unlabelled but in a machine learning context can be used to pre-train networks using self-supervised learning techniques or to quantitatively analyze the image quality of consumer-collected skin images.

Background & Summary

For over two decades, computer scientists and dermatologists have attempted to create algorithms that can detect melanoma using digital images of pigmented skin lesions. In 1987, proposals were already being made to use computers as melanoma diagnostic tools¹. In 1994, one of the earliest efforts used neural network with 240 images and claimed 80.7% correct classification². By 2012 we were already seeing reviews of computerized analysis of pigmented skin lesions³. More recent works claim accuracy on par with dermatologists with datasets that now include hundreds of thousands of images^{4–6}. In contrast, modern machine learning architectures using vision transformer backbones for image processing tasks including classification rely on hundreds of millions of images^{7,8}. The dual escalation of training dataset size and model complexity shows no signs of stopping. However, amassing medical datasets on the order of hundreds of thousands or even millions of images presents many difficulties. The cost of collection in a clinical setting, privacy issues, and research oversight by review boards are just some examples of impediments. To begin to address that challenge, MoleMapper was an IRB-approved project (#16038) designed to use smartphones in non-clinical settings to acquire a massive dataset that could be shared with a larger research community.

MoleMapper is a skin lesion tracking application for smartphones that allowed users to participate in an OHSU IRB approved research project from June 2016 through December 2022⁹. Participants used their smartphones to photograph and identify concerning moles with the ability to track changes over time. These images were securely uploaded to the hosting platform maintained by Sage Bionetworks where they were subsequently curated by the OHSU team to exclude spurious images or images with protected health identifiers.

MoleMapper was initially released as a collaboration between Sage Bionetworks, Apple, OHSU and the initial author, Dan Webster under the oversight of the Western IRB. The data from that first project have been released¹⁰. In June of 2016, OHSU took over ownership and oversight by the OHSU IRB (#16038). A subsequent software version for the Android platform was released as a collaboration between the ResearchStack development team, Sage Bionetworks and OHSU as a demonstration of the ResearchStack framework. The Android

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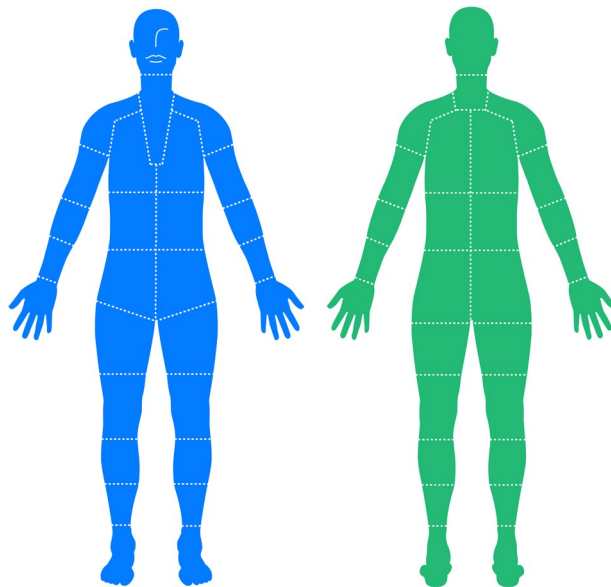


Fig. 1 MoleMapper zones. Front and back mannequins used to identify small regions associated with mole locations.

software version was only supported for one release. The iOS interface underwent a significant user-interface change between the iOS 2.x versions and the 3.x versions.

The intent of the MoleMapper study was to collect self-labelled longitudinal data of pigmented skin lesions (PSLs) by ordinary users in their various settings and to provide a useful tool for users to photographically record and track their moles. These images are in contrast to photographs taken with specialized cameras in a clinical setting by healthcare providers. Our main hypothesis was that longitudinal images of PSLs coupled with contextual skin image patches would provide a better foundation than single PSL images for building machine learning classifiers. Our secondary hypothesis was that the use of an application on a smartphone would cost-effectively improve the number of participants in this type of study by orders of magnitude. The results related to those hypotheses will be reported later in a final report.

These data fill a unique gap in the set of skin image data collections. Most publicly available skin image datasets are dermoscopic in nature (e.g. HAM10000 and ISIC). The recently released SLICE-3D dataset similarly crops lesion images, though in their case from total body photography images obtained in a clinical setting¹¹. Several image sets used to solve computer vision problems or train artificial intelligence models are proprietary in nature (e.g. MoleScope and SkinVision). Some datasets are available but at a cost. Detailed reviews of datasets and their use specifically in artificial intelligence work can be found in Wen *et al.*¹² and Melarkode *et al.*¹³ These released MoleMapper data are non-dermoscopic skin images that are freely available to the research community,

Methods

Smartphone applications. The MoleMapper application was designed for Apple iPhones using standard iOS frameworks to control the user interface and camera. The participant enrolment process used Apple's ResearchKit framework and Sage Bionetwork's Bridge Software Development Kit (now discontinued). Various third-party libraries were used to encrypt and compress the data before transmission to the Bridge servers. A version of the MoleMapper application was created for the Android platform that used standard Android application programming interfaces (APIs) and ResearchStack, an Android framework similar to ResearchKit.

Software application versions. In the iOS 2.x and the Android 1.x versions, users were asked to take a contextual photograph of a zone on the body identified using a mannequin figure (Fig. 1). Users then identified pigmented skin lesions generically called moles on the zone photograph and were then asked to take additional photographs of each individual mole, preferably including a US coin for measurement purposes. On this photograph, users were asked to place and then reposition/resize a circle using "arrow" and "+" or "-" buttons to circumscribe the mole and, if included, a reference coin. They also identified the coin denomination if included. The application had many other features, e.g. mole naming and the display of mole statistics, that we refrain from describing here because they do not pertain to the released dataset.

In the iOS 3.x versions, the photography process was changed significantly. Users were still asked to take a zone photograph and identify moles on it by tapping and using standard pinch and drag actions to resize and move circumscribing circles. Analysis of the resolution of rear-facing camera of iPhones beginning with the iPhone 6s led us to hypothesize that we could crop mole images from the zone photograph with sufficient resolution to be clinically useful and support machine learning efforts.

Enrolment. While the study did involve human subjects and cancer research, the OHSU IRB deemed it a low-risk study because no medical advice was offered and no disease intervention was involved. Consequently,



Fig. 2 Example zone images. These illustrate some of the variation with this approach to mole photography.

we were able to enroll participants digitally. The study enrollment process consisted of presenting the user with a sequence of informative screens describing the study, what was expected of them, possible risks, how the data was collected and stored, and how to withdraw from the study if desired. After presenting the informative screens, questions were asked to verify their understanding of the information. Upon correctly answering the questions, users could digitally sign a consent form. They then created an account on the Sage Bionetworks Bridge platform using their email address and a password which was stored in the application for future logins. Once they successfully created an account that was verified by an email, they were presented with a survey to collect additional data about the participant.

Prior to consent and included in the digitally signed consent document the enrollees received, potential participants were informed that their de-identified metadata obtained from surveys such as gender and history of melanoma, or from mole measurements would be made available to qualified researchers using the Sage Synapse system after supplying a use statement. They also were informed that the release of de-identified mole photographs would follow that same process.

Enrolment was limited to adults. Prior to beginning the enrolment process they were asked to confirm they were age 18 or older and, if not, thanked for being willing to participate but told they were ineligible. No other exclusion criteria were used. It should be noted that we did not enrich our enrollment for Fitzpatrick skin types IV + and consequently the data suffer from the same biases seen in other skin image data sets¹⁴.

Data de-identification. The MoleMapper application collected metadata through surveys and image data using the smartphone cameras. Data was kept private and not visible to other applications including the phone's photo library. Prior to transferring data to the back-end infrastructure provided by Sage Bionetworks, the data was securely encrypted and submitted to the account created during enrollment. There, any identifying information provided during enrollment was stripped from the data before being transferred to the Study Team's private Synapse platform. Subsequent de-identification of images, participant age, or geographic data was performed by the study team during curation.

Curation. Camera parameters and capture date are retained in the image EXIF data but GPS information has been removed. A subset of zone images have been manually chosen to represent the diversity of such images and to provide example context for cropped mole and skin patch images. Example zone images can be seen in Fig. 2. These zone images show examples of participants including coins as size proxies and not including coins. From the complete set of zone images, moles identified by the participants in their images have been cropped to increase the number of images that can be published with HIPAA compliance. These are cropped with approximately three times the width and height of the mole where possible. For instance, if a mole in a zone image was 100 pixels wide and 150 pixels tall, we would have cropped it to approximately 300 pixels by 450 pixels. We use pixels because absolute sizes could not be obtained from the smartphone camera systems, coins were not

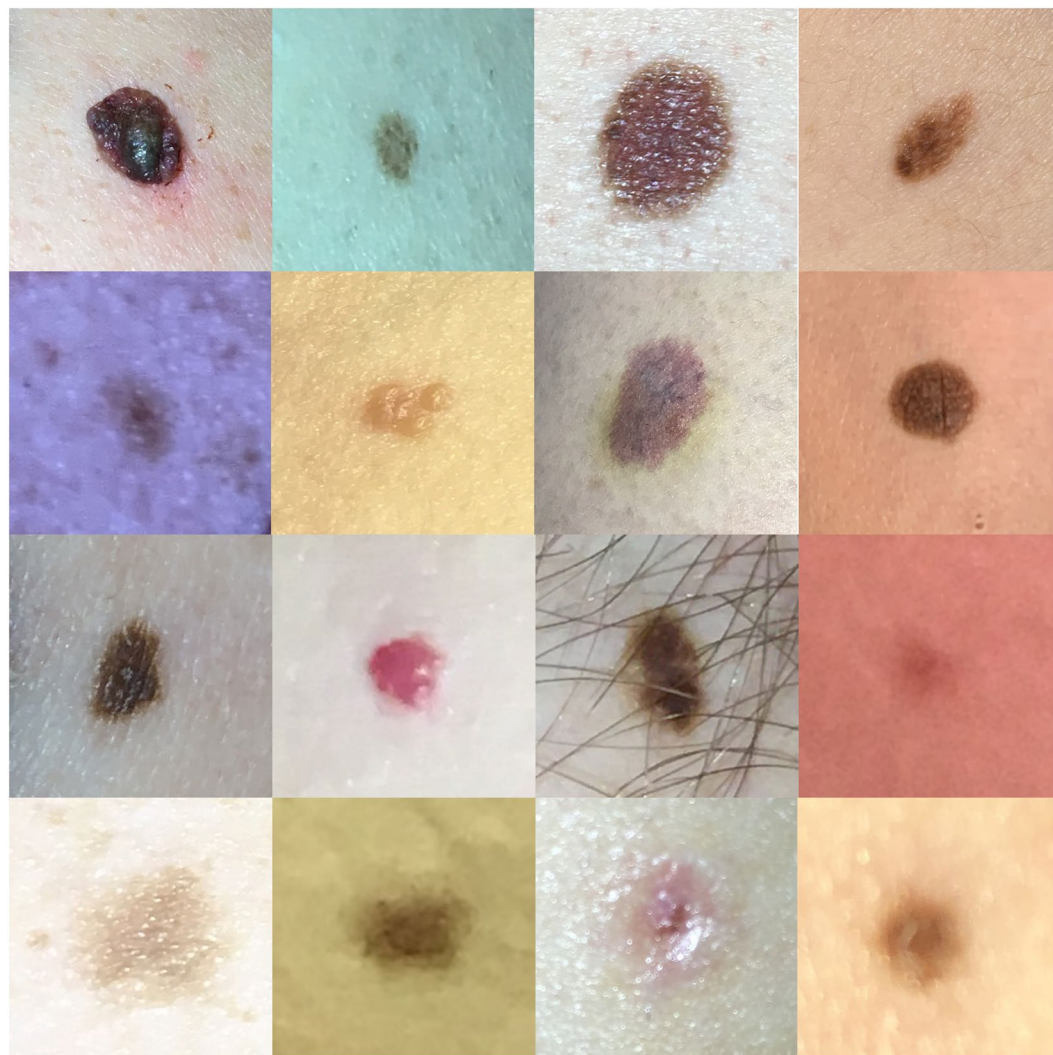


Fig. 3 Example mole images extracted from zone images. These show some of the variation seen in photographed moles as well as some of the artifacts such as blur, hair, and color problems.

universally used, and when used there remains questions about the accuracy of the calculation using a proxy. Example mole images can be seen in Fig. 3. An additional set of skin patch images were cropped from the zone images to be used in research correlating skin conditions with mole appearance. These were randomly cropped to be approximately the same size as cropped mole images. Curation was done by members of the IRB-approved study team. See Fig. 4 for an illustration of a zone images with extracted mole images. HIPAA rules related to the metadata, specifically participant age and location, were also used to filter records.

Data Records

The OHSU MoleMapper Skin Images¹⁵ data is available on the Sage Bionetworks Synapse platform accessed at <https://doi.org/10.7303/syn51520810>. Researchers seeking access to the data must be a Certified User on the Synapse platform and request access to the data by providing a use statement. This dataset includes 27,499 cropped mole images curated to exclude images with protected health identifiers, 7,305 cropped images of skin patches near the mole images, 1,000 contextual ‘zone’ images, and the following metadata from 4,158 participants: age when onboarding, gender at birth, and personal history of melanoma. Approximately 12% of moles were measured more than once. The total size for all images and metadata is less than 3.5 gigabytes.

Five tables of data comprise the dataset as seen in Fig. 5. Standard “crow’s foot” notation is used to show the relationships between the tables. For example, a participant can have multiple zone records while a zone record must belong to one and only one participant. Zones in turn may have zero or more moles and zero or mole skin patches but one zone name. Detailed schema information is provided with each table on the Synapse platform.

A subset of the participant metadata has been included with the dataset. These include their gender designated at birth, the age they were when they enrolled in the study, and whether they have a personal history of melanoma or not. Participants are uniquely identified by a semantically meaningless health_code id. Early versions of the MoleMapper application allowed participants to skip questions that were made mandatory later. Likewise, early versions allowed users to specify ‘other’ as their birth gender and this was restricted later. Of

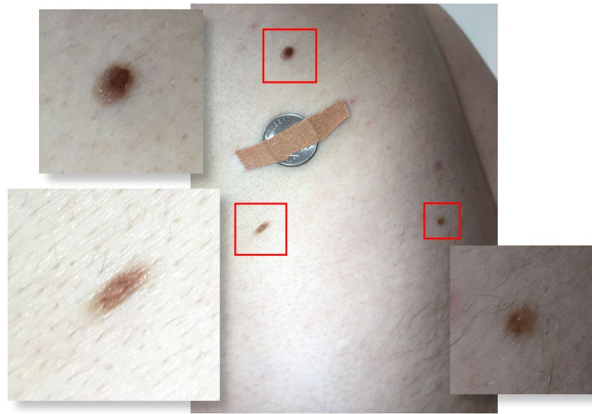


Fig. 4 Composite image showing relationship between individual moles and a zone image.

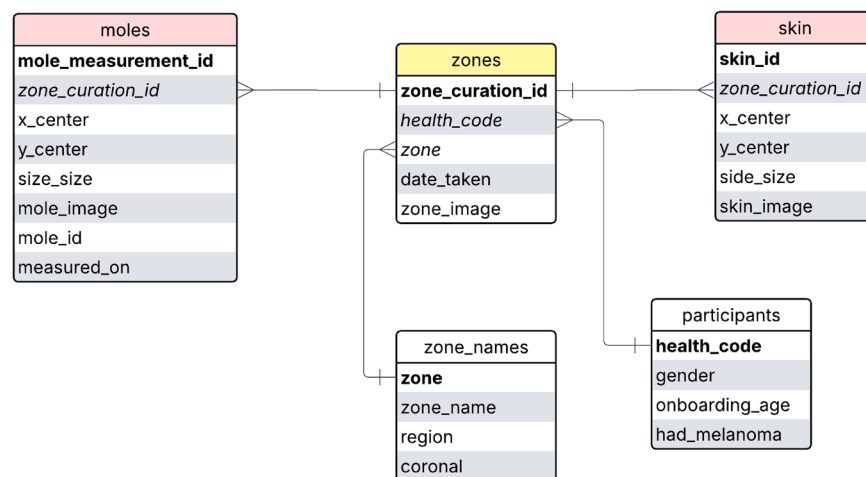


Fig. 5 Entity Relationship Diagram of the data. This shows, for example, how zone images can have multiple mole and skin patch images and how participants can have multiple zone images. Moles that have been remeasured can be identified using the mole_id and measured_on attributes in the mole table.

those who provided metadata, 2,426 identified as ‘female’, 1673 identified as ‘male’ and 58 identified as ‘other’. For the melanoma question, 3,501 had a history of melanoma and 656 did not. These numbers reflect the data in this dataset and do not include those from the earlier reported dataset¹⁰. See Fig. 6 for plots illustrating these numbers.

A zone record is connected to a participant by the health_code. It is uniquely identified by a zone_curation_id and contains a zone number, the date taken and a pointer to the image which can be downloaded. Records for zones that are the source of mole and skin images are included but not all zones have their source images. This was because some zone images may contain personal health information (PHI) while the cropped mole(s) do not. We also were constrained by storage limitations and chose to limit the number of zone images. We have excluded the 2.x (iOS) images because they were captured at phone display resolution and are deemed to be of insufficient resolution for the research envisioned for this dataset.

Mole images were created by cropping an image region around the participant-identified mole from the zone or pseudo-zone context image. Mole location and size information (in pixels) is included so that when a zone image exists for the parent zone, researchers can analyze accuracy and variation in participants’ encircling efforts. Examples can be seen in Fig. 3.

Skin patches were cropped by the curators to provide additional context. This is potentially useful in looking for sun damage in skin near moles that were of concern to the participants. Location and size information is again included for those cases where zone images exist if researchers want to understand more about the relationship between the patch image and the mole image.

A lookup table translating zone numbers into zone names is also provided. Zones have been further grouped into regions and identified by coronal orientation (anterior or posterior). Regions include names like ‘Trunk’, ‘Arm’, ‘Hand’ and ‘Back’. Zone names are more specific; some examples of zone names are ‘Right Upper Arm’, ‘Left Upper Back’ and ‘Right Hand’.

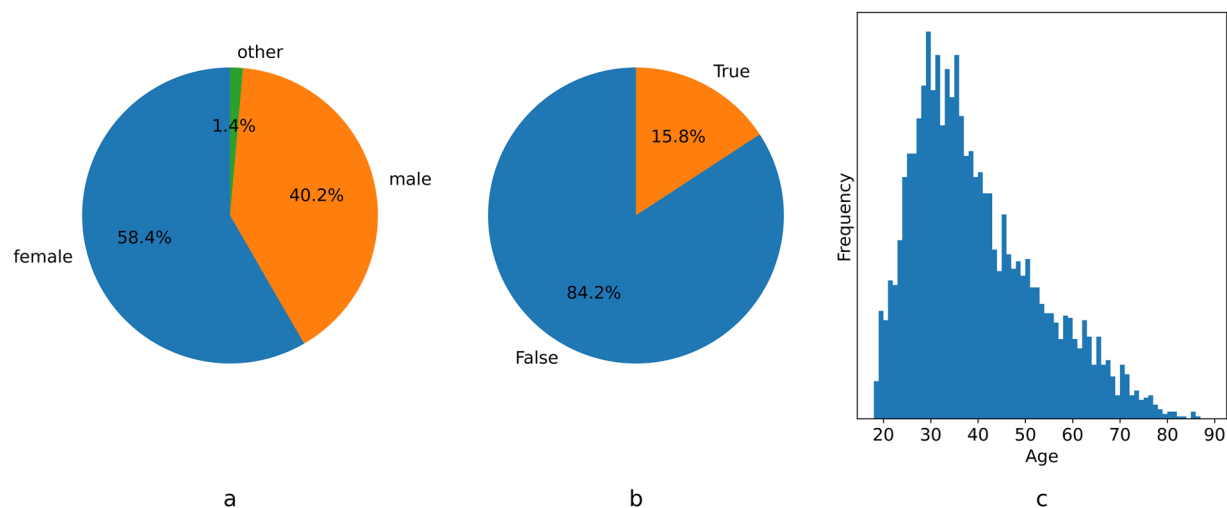


Fig. 6 Metadata plots. The left pie chart (a) shows the participant gender distribution, the middle pie-chart (b) shows the participant history of melanoma distribution, and the right histogram (c) shows the distribution of participant ages.

Technical Validation

One of the purposes of this project was to observe “in the wild” photography of skin lesions by the general public using smartphone devices. Vendor-provided APIs for controlling the cameras were used as per their documentation. Images were not manipulated either during acquisition or after. Standard software quality assurance testing was performed by a test team prior to each software version release.

Prior to publishing, images were filtered based on HIPAA compliance rules.

Usage Notes

To use these data, researchers must create an account on Sage Bionetwork’s Synapse platform and become a Certified User. Once certified, they may request access to the OHSU MoleMapper project (syn51520810) by providing a use statement.

The Synapse platform contains documentation on how to become a Certified User, how to view and download data and can be accessed at <https://help.synapse.org>.

Code availability

MoleMapper is an open source project and the code is available at https://github.com/ohsu-molemapper/MoleMapper_Final. Help and example code to retrieve the data using the command line, Python, R or REST methods are available on the Synapse.org website. The Python code used to curate the images by filtering out test users and images with patient health information and to extract mole and skin patches contains internally sensitive information such as email account data. The processing is rudimentary and did not manipulate the data other than cropping and we are therefore not releasing it publicly.

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Author contributions

Tracy Petrie was the program manager for the MoleMapper project and oversaw software development and ongoing data analysis. Ravikant Samatham contributed to the software development effort and was the principal curator for the images. Daniel E. Webster was the original author of the MoleMapper application and has provided invaluable assistance in preparing this manuscript. Sancy A. Leachman gave significant editorial input in preparing this manuscript.

Competing interests

Daniel E. Webster is currently an employee of AbbVie Inc. and may own AbbVie stock.

Additional information

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